<!DOCTYPE html> <html lang="uz"> <head> <meta charset="UTF-8"> <meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>9-Sinf Fizika: Modda tuzilishining molekulyar-kinetik nazariyasi</title> <!-- Tailwind CSS CDN --> <script src="https://cdn.tailwindcss.com"></script> <!-- MathJax for LaTeX formulas --> <script src="https://polyfill.io/v3/polyfill.min.js?features=es6"></script> <script id="MathJax-script" async src="https://cdn.jsdelivr.net/npm/mathjax@3/es5/tex-mml-chtml.js"></script> <style> @import url('https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap'); body { font-family: 'Inter', sans-serif; } </style> </head> <body class="bg-slate-50 text-slate-800 antialiased min-h-screen flex flex-col">
<!-- Header --> <header class="bg-indigo-600 text-white shadow-lg py-6 px-4 mb-8"> 9-Sinf Fizika Fani <h1 class="text-3xl font-bold mt-1">1-mavzu: Modda tuzilishining molekulyar-kinetik nazariyasi</h1> Nazariy ma'lumotlar, tajribiy qoidalar va vizual simulyatsiya ssenariylari </header>
<!-- Main Container --> <main class="max-w-4xl mx-auto px-4 flex-grow space-y-8 mb-12">
<!-- Plan Overview Box --> <section class="bg-white rounded-xl shadow-sm border border-slate-200 p-6"> <h2 class="text-xl font-semibold text-indigo-700 mb-4 flex items-center gap-2"> <svg class="w-5 h-5" fill="none" stroke="currentColor" viewBox="0 0 24 24"><path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M9 5H7a2 2 0 00-2 2v12a2 2 0 002 2h10a2 2 0 002-2V7a2 2 0 00-2-2h-2M9 5a2 2 0 002 2h2a2 2 0 002-2M9 5a2 2 0 012-2h2a2 2 0 012-2m-6 9l2 2 4-4"></path></svg> Dars rejasi </h2> <ul class="grid md:grid-cols-3 gap-4 text-sm font-medium"> <li class="bg-indigo-50 p-3 rounded-lg text-indigo-900 border border-indigo-100">1. Molekulyar-kinetik nazariya (MKT) <li class="bg-indigo-50 p-3 rounded-lg text-indigo-900 border border-indigo-100">2. MKTning tajribalarda isbotlangan 3 ta qoidasi <li class="bg-indigo-50 p-3 rounded-lg text-indigo-900 border border-indigo-100">3. Braun harakati va uning ahamiyati </section>
<!-- Section 1 --> <section class="bg-white rounded-xl shadow-sm border border-slate-200 p-6 md:p-8 space-y-6"> 1-REJA <h2 class="text-2xl font-bold text-slate-900 mt-2">Molekulyar-kinetik nazariya (MKT)</h2>   <h3 class="text-lg font-semibold text-slate-800 mb-2">📚 Nazariy ma’lumot</h3> Molekulyar-kinetik nazariya (MKT) — moddalarning tuzilishi, xossalari va issiqlik jarayonlarini ularni tashkil etuvchi eng kichik zarrachalar (atom va molekulalar)ning tartibsiz mexanik harakati hamda o‘zaro ta’siri nuqtai nazaridan tushuntiradigan fundamental ta’limotdir.   Tarixi: Bu nazariyaning ilk asoslari qadimgi mutafakkirlar (Demokrit, Lukretsiy) tomonidan ilgari surilgan bo‘lib, XIX asrning ikkinchi yarmida M. Lomonosov, J. Maksvell, L. Bolsman va boshqa olimlar tomonidan ilmiy jihatdan to‘laqonli isbotlangan va rivojlantirilgan.
 <h4 class="text-md font-semibold text-slate-800 mb-3 flex items-center gap-2"> <svg class="w-4 h-4 text-indigo-600" fill="none" stroke="currentColor" viewBox="0 0 24 24"><path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M15 10l4.553-2.276A1 1 0 0121 8.618v6.764a1 1 0 01-1.447.894L15 14M5 18h8a2 2 0 002-2V8a2 2 0 00-2-2H5a2 2 0 00-2 2v8a2 2 0 002 2z"></path></svg> Simulyatsiya va animatsiya ssenariysi </h4> Raqamli simulyatsiya (PhET tavsiyasi): “States of Matter: Basics” (Moddalarning agregat holatlari)

simulyatsiyasi. Bunda o‘quvchilar gaz, suyuqlik va qattiq jism idishdagi molekulalarini ko‘rib, ularning harakat tezligini haroratni o‘zgartirish orqali kuzatadilar (T \uparrow \rightarrow v \uparrow). Vizual animatsiya ssenariysi: Ekranda suv tomchisi ko‘rsatiladi. Kamera asta-sekin yaqinlashib (zoom effekt), tomchi ichidagi milliardlab mayda, ko‘zga ko‘rinmas zarrachalar — molekulalarga bo‘linganini namoyish etadi. Zarrachalar doimiy ravishda bir-biriga urilib, tebranib turgani vizual tarzda yorqin ranglarda jonlantiriladi.   </section>
<!-- Section 2 --> <section class="bg-white rounded-xl shadow-sm border border-slate-200 p-6 md:p-8 space-y-6"> 2-REJA <h2 class="text-2xl font-bold text-slate-900 mt-2">MKTning tajribalarda isbotlangan uchta qoidasi</h2>
 <h3 class="text-lg font-semibold text-slate-800 mb-3">📚 Nazariy ma’lumot</h3> MKT quyidagi fundamental qoidalarga tayanadi va ularning har biri hayotiy tajribalar orqali to‘la isbotlangan:
  <h4 class="font-semibold text-indigo-900 mb-1">1. Moddalar diskret (uzlukli) tuzilishga ega</h4> Har qanday jism yaxlit ko‘rinsada, aslida juda kichik alohida zarrachalar — molekula va atomlardan tashkil topgan. Zarrachalar orasida bo‘shliqlar mavjud.   <h4 class="font-semibold text-indigo-900 mb-1">2. Molekulalar tinimsiz va tartibsiz harakatda bo‘ladi</h4> Bu harakat issiqlik harakati deb ataladi, chunki uning tezligi jismning haroratiga bog‘liq (harorat qancha baland bo‘lsa, zarrachalar shuncha tez harakat qiladi).   <h4 class="font-semibold text-indigo-900 mb-1">3. Molekulalar orasida o‘zaro ta’sir kuchlari mavjud</h4> Zarrachalar bir vaqtning o‘zida ham tortishish, ham itarishish kuchlarini namoyon qiladi. Ularning qaysisi ustun kelishi zarrachalar orasidagi masofaga bog‘liq.
 <h4 class="text-md font-semibold text-slate-800 mb-3 flex items-center gap-2"> <svg class="w-4 h-4 text-indigo-600" fill="none" stroke="currentColor" viewBox="0 0 24 24"><path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M15 10l4.553-2.276A1 1 0 0121 8.618v6.764a1 1 0 01-1.447.894L15 14M5 18h8a2 2 0 002-2V8a2 2 0 00-2-2H5a2 2 0 00-2 2v8a2 2 0 002 2z"></path></svg> Simulyatsiya va animatsiya ssenariysi </h4> Raqamli simulyatsiya: Ikki molekula orasidagi masofani o‘zgartirganda itarish va tortish kuchlarining o‘zgarish grafigini ko‘rsatuvchi interaktiv dastur (masalan, molekulyar o‘zaro ta'sir modeli). Vizual animatsiya ssenariysi: • 1-sahna: Qand bo‘lagining suvda erish jarayoni tezlashtirilgan animatsiyada ko‘rsatiladi (diskret tuzilish isboti). • 2-sahna: Ikki molekula bir-biriga yaqinlashganda repulsiv (itarish) kuchlari, uzoqlashganda esa tortishish kuchi ularni tortib turgani mexanik yaylar misolida ko‘rsatib beriladi.   </section>
<!-- Section 3 --> <section class="bg-white rounded-xl shadow-sm border border-slate-200 p-6 md:p-8 space-y-6"> 3-REJA <h2 class="text-2xl font-bold text-slate-900 mt-2">Braun harakati</h2>
 <h3 class="text-lg font-semibold text-slate-800 mb-2">📚 Nazariy ma’lumot</h3> <ul

class="space-y-3 text-slate-600"> <li class="flex items-start gap-2"> ✓ Ta’rifi va kashf etilishi: Suyuqlik yoki gazda muallaq turgan qattiq zarrachalarning suyuqlik (yoki gaz) molekulalari bilan to‘qnashishi tufayli bajaradigan tinimsiz, tartibsiz harakati Braun harakati deyiladi. Uni 1827-yilda ingliz botaniki Robert Braun mikroskop ostida gul changining suv ichidagi harakatini kuzatish davomida kashf etgan. <li class="flex items-start gap-2"> ✓ Kelib chiqish sababi: Muallaq zarrachaning har tarafdan suyuqlik molekulalari tomonidan bir xilda bo‘lmagan (tartibsiz) zarbalar olishi natijasida zarrachaning trayektoriyasi sinchkovlik bilan o‘zgarib, zig-zag shaklida sakrab yurishi. <li class="flex items-start gap-2"> ✓ Ahamiyati: Braun harakati molekulyar-kinetik nazariyaning eng yorqin tajribaviy dalillaridan biri hisoblanadi. U molekulalarning haqiqatan ham tinimsiz harakatda ekanligini ko‘z bilan ko‘rsatib beradi.
 <h4 class="text-md font-semibold text-slate-800 mb-3 flex items-center gap-2"> <svg class="w-4 h-4 text-indigo-600" fill="none" stroke="currentColor" viewBox="0 0 24 24"><path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M15 10l4.553-2.276A1 1 0 0121 8.618v6.764a1 1 0 01-1.447.894L15 14M5 18h8a2 2 0 002-2V8a2 2 0 00-2-2H5a2 2 0 00-2 2v8a2 2 0 002 2z"></path></svg> Simulyatsiya va animatsiya ssenariysi </h4> Raqamli simulyatsiya: Mayda chang zarrachasi va uni har tomondan tinimsiz urib turgan yuzlab kichik sharchalar (molekulalar) harakatini ko‘rsatuvchi PhET yoki o‘xshash fizika simulyatsiyasi. Vizual animatsiya ssenariysi: Mikroskop ostidagi ko‘rinish. Ekranning markazida yirikroq zarracha (gul changi) turibdi. Uni har tomondan mayda, tez harakatlanuvchi nuqtachalar (suv molekulalari) tinimsiz urib turibdi. Natijada markaziy zarrachaning tartibsiz, to‘satdan yo‘nalishini o‘zgartirib sakrab harakatlanishi chizig‘i (zig-zag yo‘li) ekranda yorqin iz qoldirib boradi.   </section>
</main>
<!-- Footer --> <footer class="bg-slate-800 text-slate-400 py-6 text-center text-sm border-t border-slate-700"> 9-sinf fizika fani dars materiallari va interaktiv simulyatsiya stsenariylari. </footer>
</body> </html>